

Chapter 21

Genomes and Their Evolution

PowerPoint® Lecture Presentations for

Biology

Eighth Edition

Neil Campbell and Jane Reece

Lectures by Chris Romero, updated by Erin Barley with contributions from Joan Sharp

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Overview: Reading the Leaves from the Tree of Life

- Complete genome sequences exist for a human, chimpanzee, *E. coli*, brewer's yeast, nematode, fruit fly, house mouse, rhesus macaque, and other organisms
- Comparisons of genomes among organisms provide information about the evolutionary history of genes and taxonomic groups

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- **Genomics** is the study of whole sets of genes and their interactions
 - **Bioinformatics** is the application of computational methods to the storage and analysis of biological data

Concept 21.1: New approaches have accelerated the pace of genome sequencing

- The most ambitious mapping project to date has been the sequencing of the human genome
- Officially begun as the **Human Genome Project** in 1990, the sequencing was largely completed by 2003
- The project had three stages:
 - Genetic (or linkage) mapping
 - Physical mapping
 - DNA sequencing

Three-Stage Approach to Genome Sequencing

- A **linkage map** (genetic map) maps the location of several thousand genetic markers on each chromosome
- A genetic marker is a gene or other identifiable DNA sequence
- Recombination frequencies are used to determine the order and relative distances between genetic markers

Cytogenetic map

Chromosome bands

Genes located
by FISH



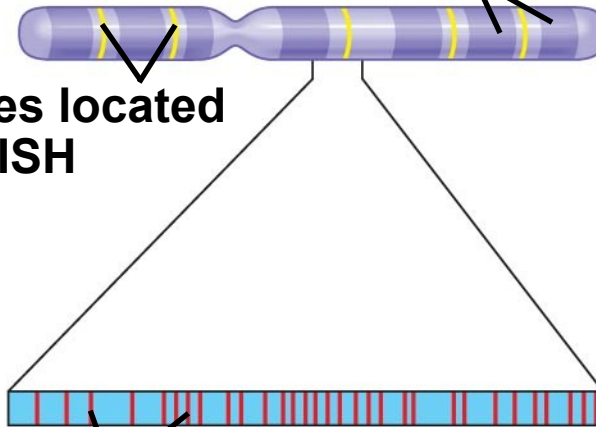
Cytogenetic map

Chromosome
bands

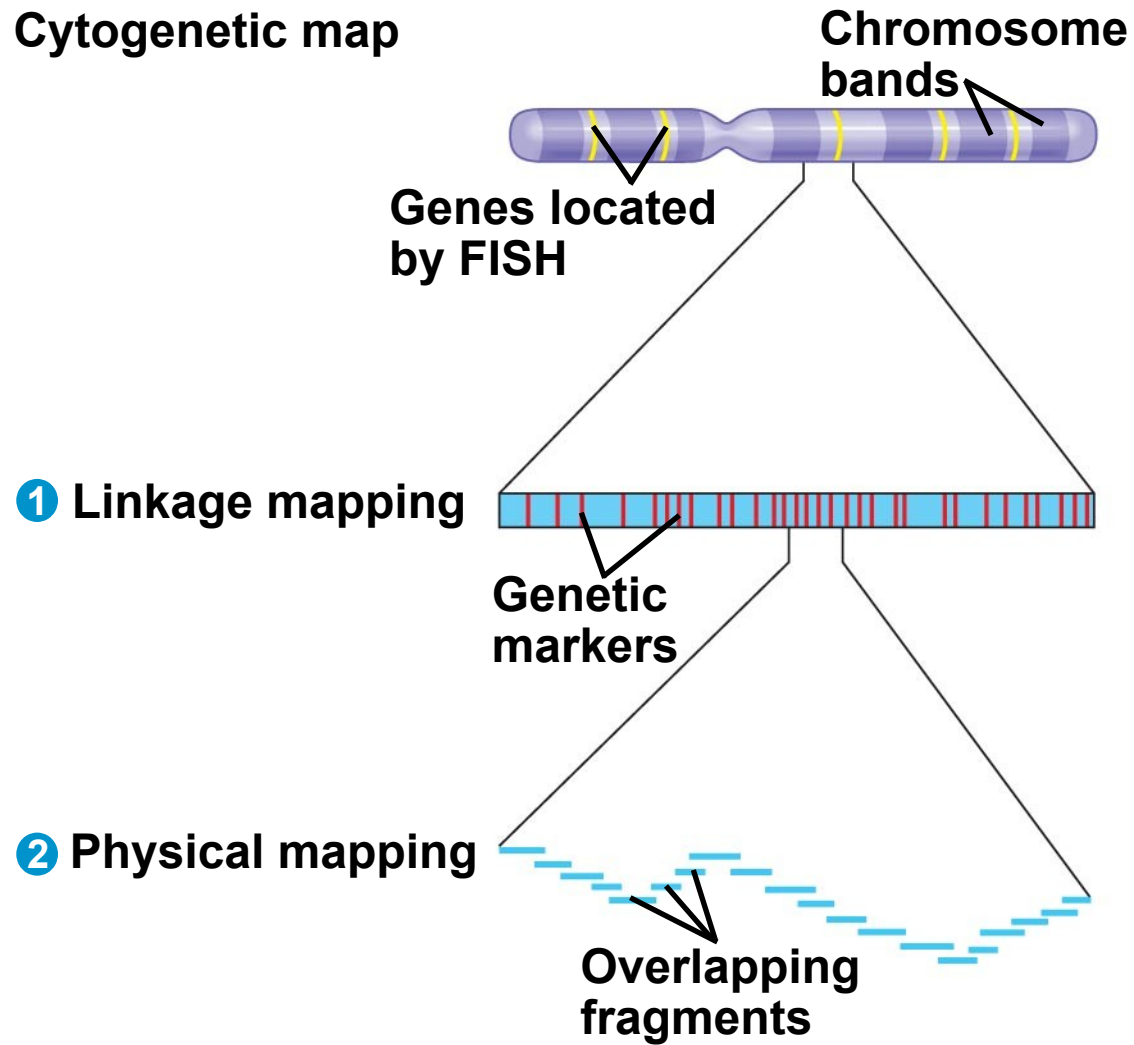
Genes located
by FISH

1 Linkage mapping

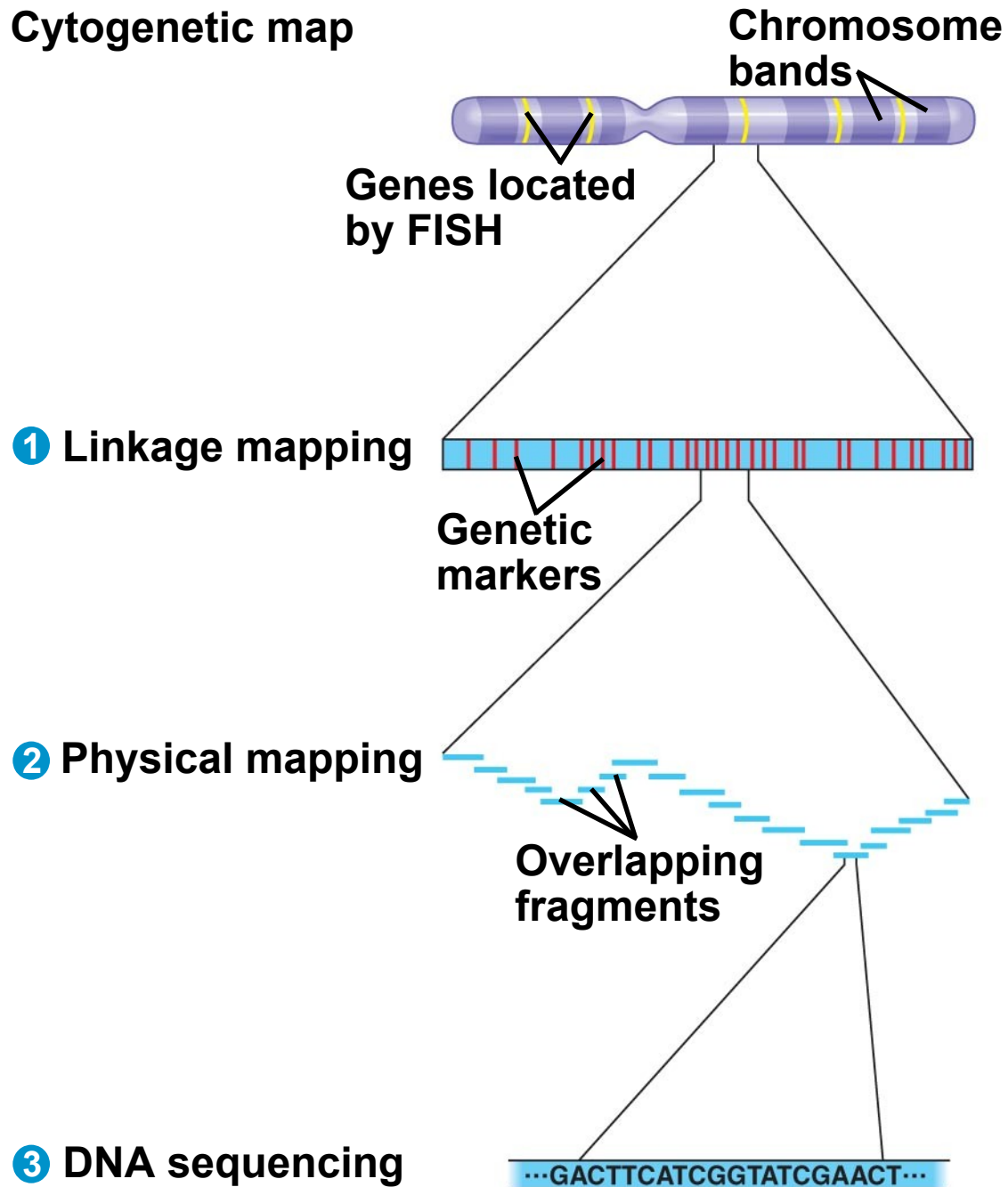
Genetic
markers



Cytogenetic map



Cytogenetic map



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- A **physical map** expresses the distance between genetic markers, usually as the number of base pairs along the DNA
 - It is constructed by cutting a DNA molecule into many short fragments and arranging them in order by identifying overlaps

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- Sequencing machines are used to determine the complete nucleotide sequence of each chromosome
 - A complete haploid set of human chromosomes consists of 3.2 billion base pairs

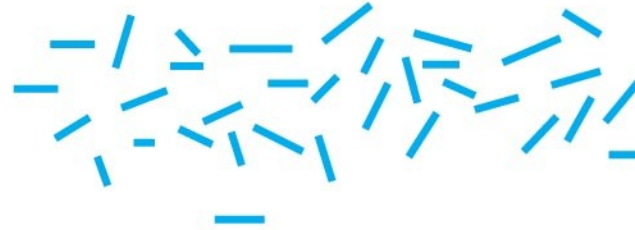
Whole-Genome Shotgun Approach to Genome Sequencing

- The whole-genome shotgun approach was developed by J. Craig Venter in 1992
- This approach skips genetic and physical mapping and sequences random DNA fragments directly
- Powerful computer programs are used to order fragments into a continuous sequence

- 1 Cut the DNA into overlapping fragments short enough for sequencing



- 2 Clone the fragments in plasmid or phage vectors.



- 1 Cut the DNA into overlapping fragments short enough for sequencing



- 2 Clone the fragments in plasmid or phage vectors.



- 3 Sequence each fragment.

CGCCATCAGT

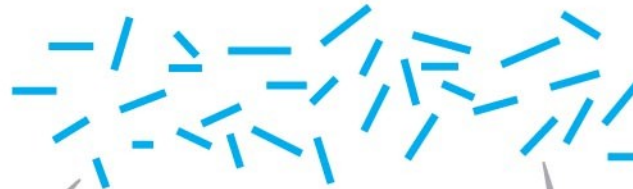
AGTCCGCTATACGA

ACGATACTGGT

- 1 Cut the DNA into overlapping fragments short enough for sequencing



- 2 Clone the fragments in plasmid or phage vectors.



- 3 Sequence each fragment.

CGCCATCAGT

AGTCCGCTATACGA

ACGATACTGGT

- 4 Order the sequences into one overall sequence with computer software.

CGCCATCAGT

ACGATACTGGT

AGTCCGCTATACGA

...CGCCATCAGTCCGCTATACGATACTGGT...

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- Both the three-stage process and the whole-genome shotgun approach were used for the Human Genome Project and for genome sequencing of other organisms
 - At first many scientists were skeptical about the whole-genome shotgun approach, but it is now widely used as the sequencing method of choice
 - A hybrid of the two approaches may be the most useful in the long run

Concept 21.2 Scientists use bioinformatics to analyze genomes and their functions

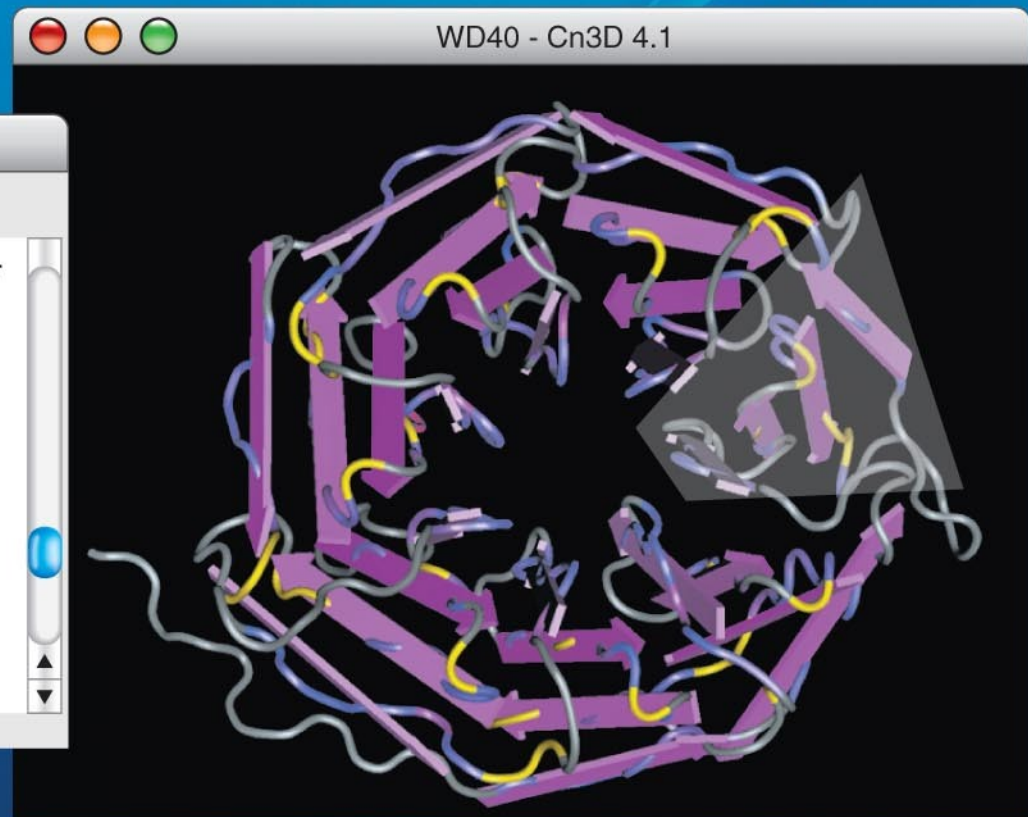
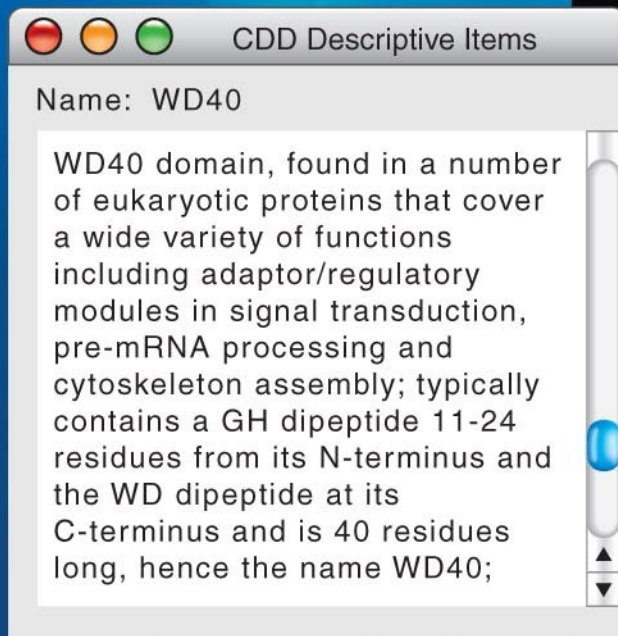
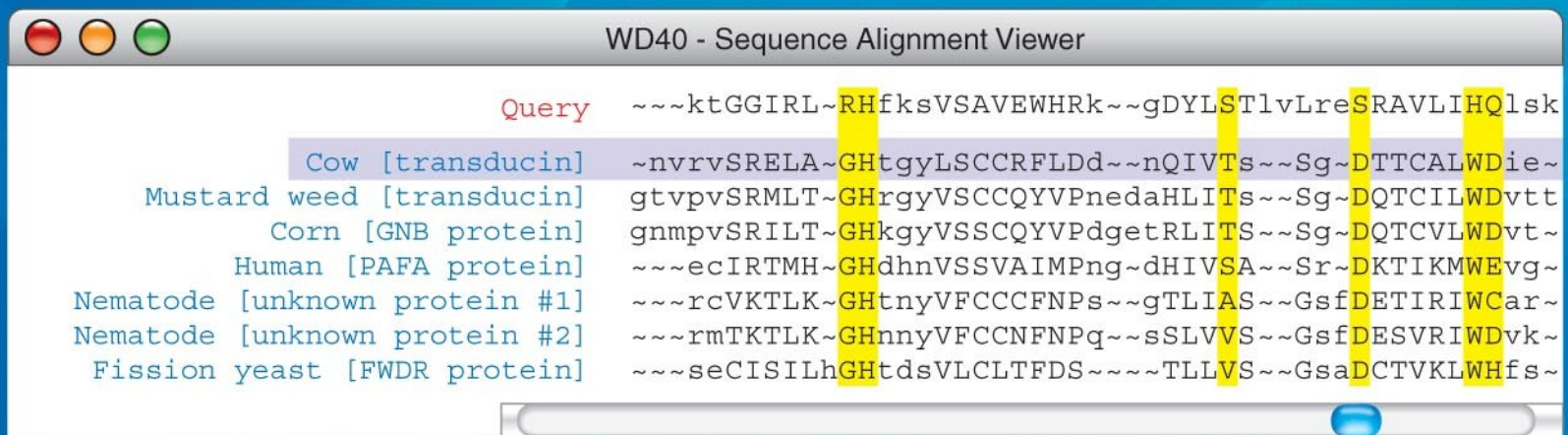
- The Human Genome Project established databases and refined analytical software to make data available on the Internet
- This has accelerated progress in DNA sequence analysis

Centralized Resources for Analyzing Genome Sequences

- Bioinformatics resources are provided by a number of sources:
 - National Library of Medicine and the National Institutes of Health (NIH) created the National Center for Biotechnology Information (NCBI)
 - European Molecular Biology Laboratory
 - DNA Data Bank of Japan

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- Genbank, the NCBI database of sequences, doubles its data approximately every 18 months
 - Software is available that allows online visitors to search Genbank for matches to:
 - A specific DNA sequence
 - A predicted protein sequence
 - Common stretches of amino acids in a protein
 - The NCBI website also provides 3-D views of all protein structures that have been determined

Fig. 21-4



Identifying Protein-Coding Genes Within DNA Sequences

- Computer analysis of genome sequences helps identify sequences likely to encode proteins
- Comparison of sequences of “new” genes with those of known genes in other species may help identify new genes

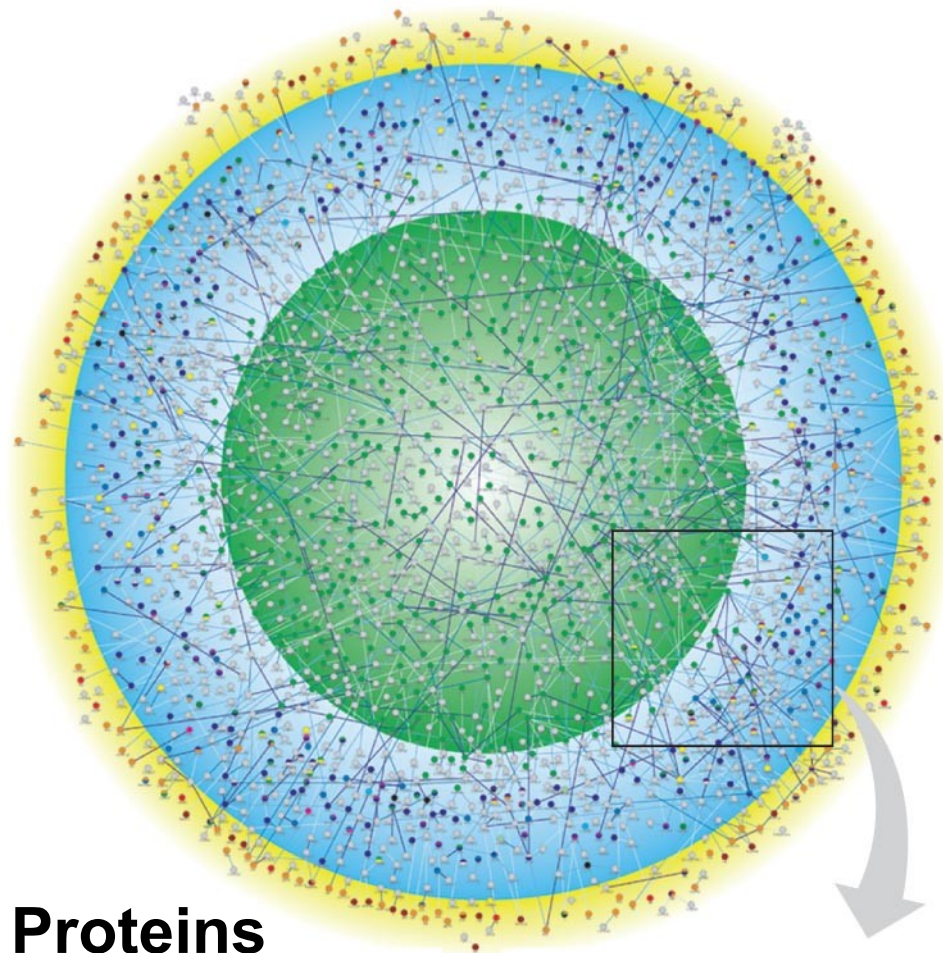
Understanding Genes and Their Products at the Systems Level

- **Proteomics** is the systematic study of all proteins encoded by a genome
- Proteins, not genes, carry out most of the activities of the cell

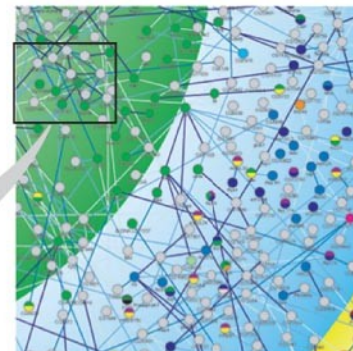
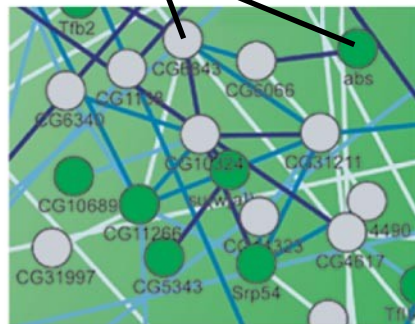
How Systems Are Studied: An Example

- A systems biology approach can be applied to define gene circuits and protein interaction networks
- Researchers working on *Drosophila* used powerful computers and software to predict 4,700 protein products that participated in 4,000 interactions
- The systems biology approach is possible because of advances in bioinformatics

Fig. 21-5



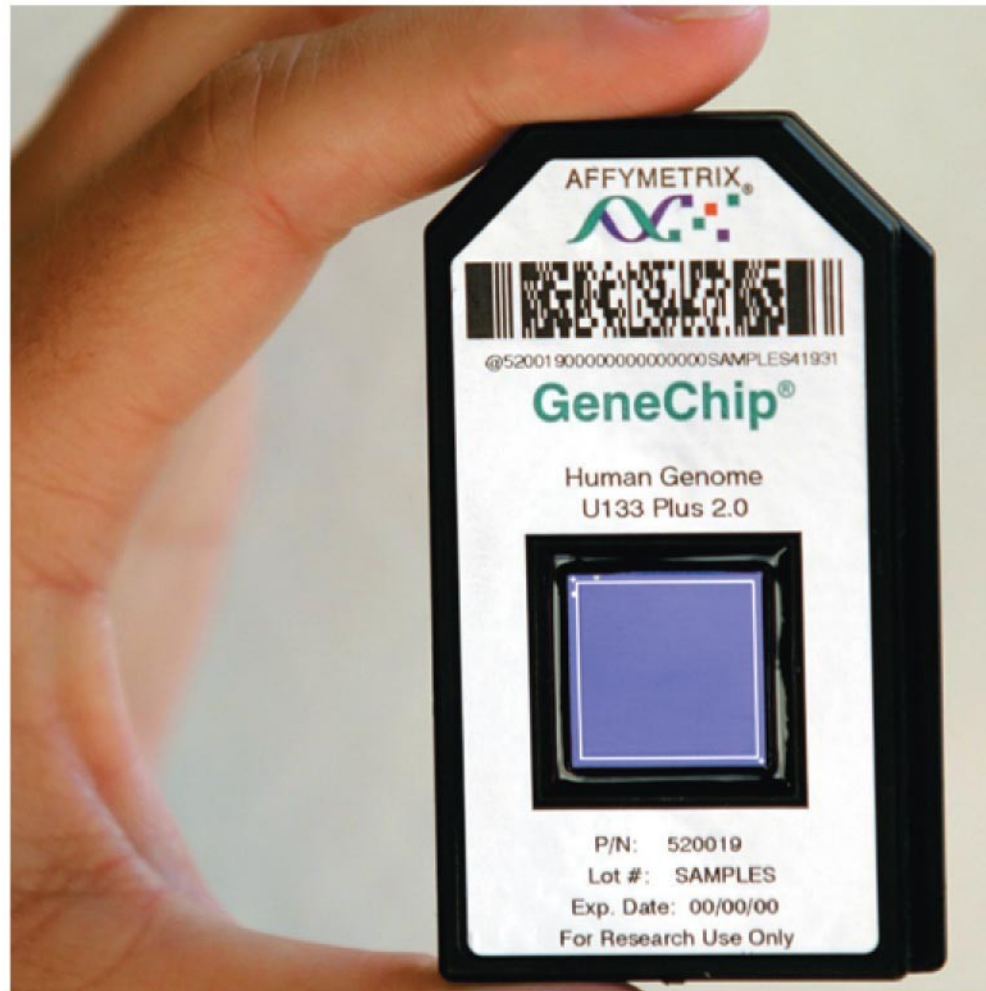
Proteins



Application of Systems Biology to Medicine

- A systems biology approach has several medical applications:
 - The Cancer Genome Atlas project is currently monitoring 2,000 genes in cancer cells for changes due to mutations and rearrangements
 - Treatment of cancers and other diseases can be individually tailored following analysis of gene expression patterns in a patient
 - In future, DNA sequencing may highlight diseases to which an individual is predisposed

Fig. 21-6



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Concept 21.3 Genomes vary in size, number of genes, and gene density

- By summer 2007, genomes had been sequenced for 500 bacteria, 45 archaea, and 65 eukaryotes including vertebrates, invertebrates, and plants

Genome Size

- Genomes of most bacteria and archaea range from 1 to 6 million base pairs (Mb); genomes of eukaryotes are usually larger
- Most plants and animals have genomes greater than 100 Mb; humans have 3,200 Mb
- Within each domain there is no systematic relationship between genome size and phenotype

Table 21.1 Genome Sizes and Estimated Numbers of Genes*

Organism	Haploid Genome Size (Mb)	Number of Genes	Genes per Mb
Bacteria			
<i>Haemophilus influenzae</i>	1.8	1,700	940
<i>Escherichia coli</i>	4.6	4,400	950
Archaea			
<i>Archaeoglobus fulgidus</i>	2.2	2,500	1,130
<i>Methanosarcina barkeri</i>	4.8	3,600	750
Eukaryotes			
<i>Saccharomyces cerevisiae</i> (yeast)	13	6,200	480
<i>Caenorhabditis elegans</i> (nematode)	100	20,000	200
<i>Arabidopsis thaliana</i> (plant)	118	25,500	215
<i>Drosophila melanogaster</i> (fruit fly)	180	13,700	76
<i>Oryza sativa</i> (rice)	390	40,000	140
<i>Danio rerio</i> (zebrafish)	1,700	23,000	13
<i>Mus musculus</i> (house mouse)	2,600	22,000	11
<i>Homo sapiens</i> (human)	3,200	20,500	7
<i>Fritillaria assyriaca</i> (plant)	120,000	ND	ND
*Some values given here are likely to be revised as genome analysis continues. Mb = million base pairs. ND = not determined.			

Number of Genes

- Free-living bacteria and archaea have 1,500 to 7,500 genes
- Unicellular fungi have from about 5,000 genes and multicellular eukaryotes from 40,000 genes

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- Number of genes is not correlated to genome size
 - For example, it is estimated that the nematode *C. elegans* has 100 Mb and 20,000 genes, while humans have 3,200 Mb and 20,488 genes
 - Vertebrate genomes can produce more than one polypeptide per gene because of alternative splicing of RNA transcripts

Gene Density and Noncoding DNA

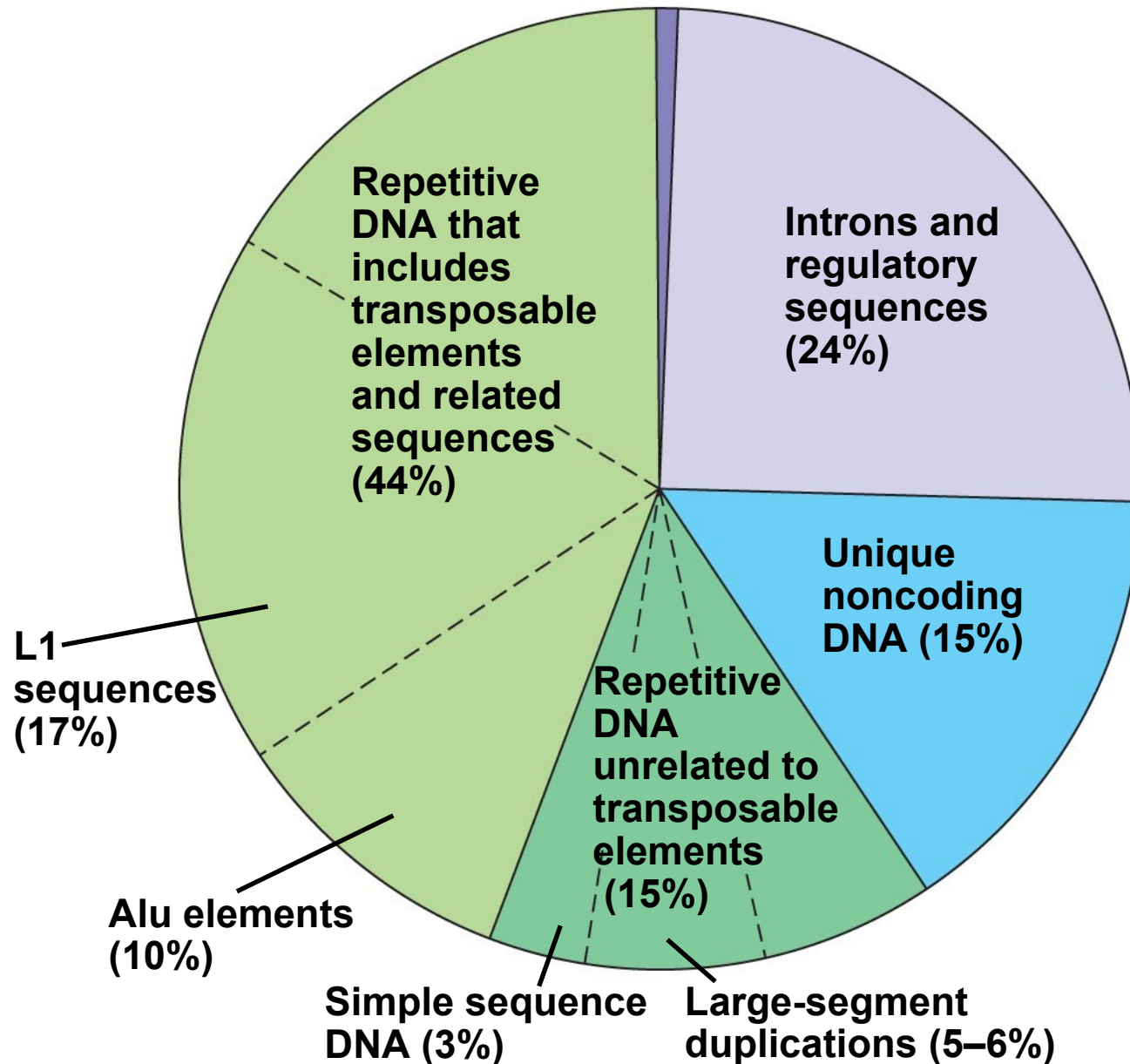
- Humans and other mammals have the lowest gene density, or number of genes, in a given length of DNA
- Multicellular eukaryotes have many introns within genes and noncoding DNA between genes

Concept 21.4: Multicellular eukaryotes have much noncoding DNA and many multigene families

- The bulk of most eukaryotic genomes consists of noncoding DNA sequences, often described in the past as “junk DNA”
- Much evidence indicates that noncoding DNA plays important roles in the cell
- For example, genomes of humans, rats, and mice show high sequence conservation for about 500 noncoding regions
- Sequencing of the human genome reveals that 98.5% does not code for proteins, rRNAs, or tRNAs

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- About 24% of the human genome codes for introns and gene-related regulatory sequences
 - Intergenic DNA is noncoding DNA found between genes
 - **Pseudogenes** are former genes that have accumulated mutations and are nonfunctional
 - **Repetitive DNA** is present in multiple copies in the genome
 - About three-fourths of repetitive DNA is made up of transposable elements and sequences related to them

Exons (regions of genes coding for protein or giving rise to rRNA or tRNA) (1.5%)



Transposable Elements and Related Sequences

- The first evidence for wandering DNA segments came from geneticist Barbara McClintock's breeding experiments with Indian corn
- McClintock identified changes in the color of corn kernels that made sense only by postulating that some genetic elements move from other genome locations into the genes for kernel color
- These **transposable elements** move from one site to another in a cell's DNA; they are present in both prokaryotes and eukaryotes

Fig. 21-8

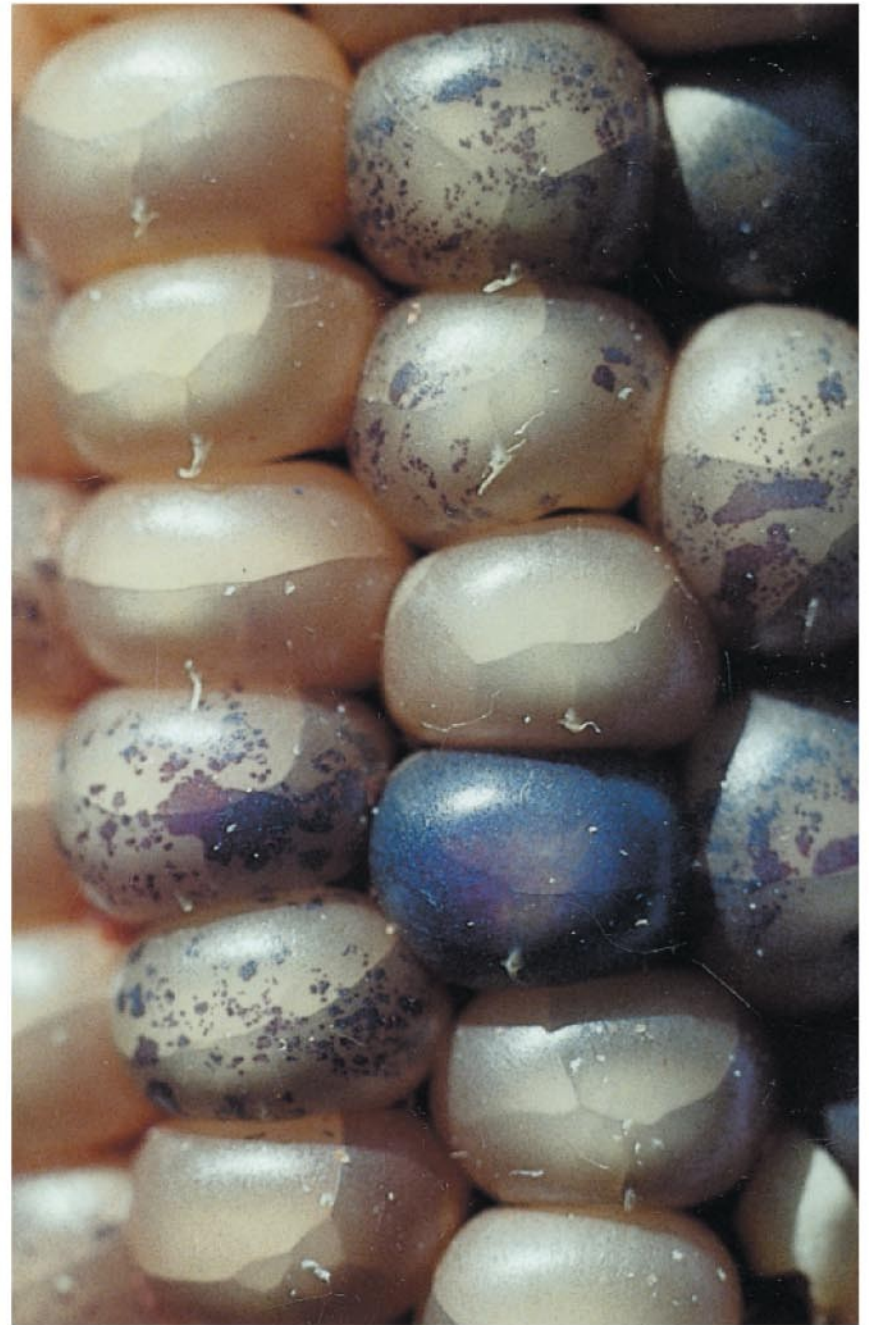


Fig. 21-8a

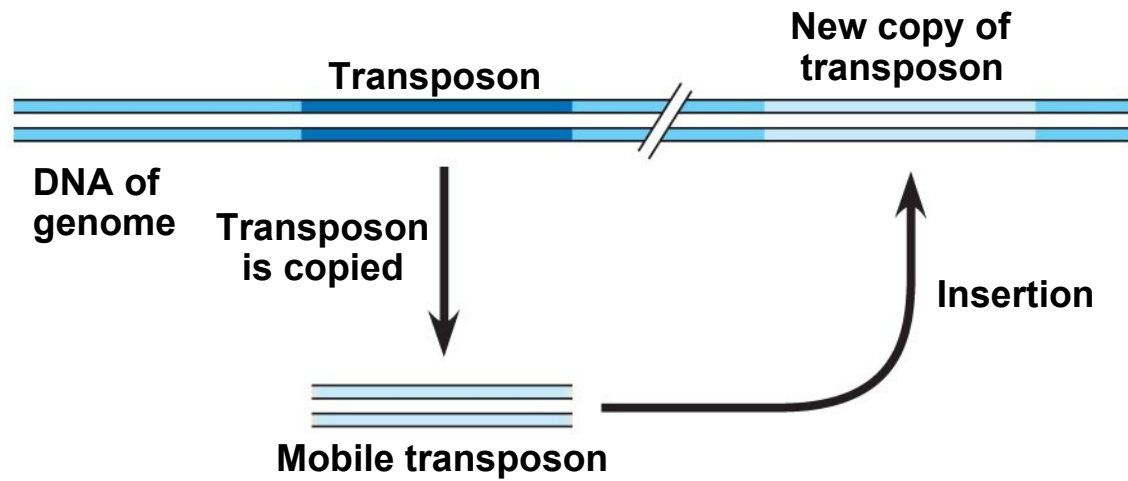


Fig. 21-8b

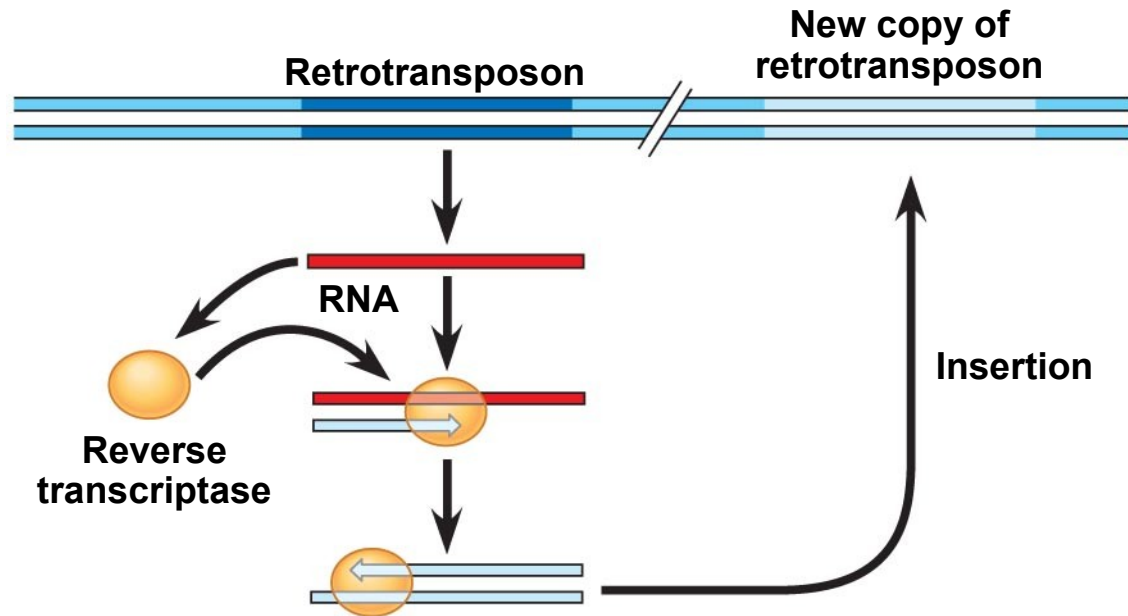


Movement of Transposons and Retrotransposons

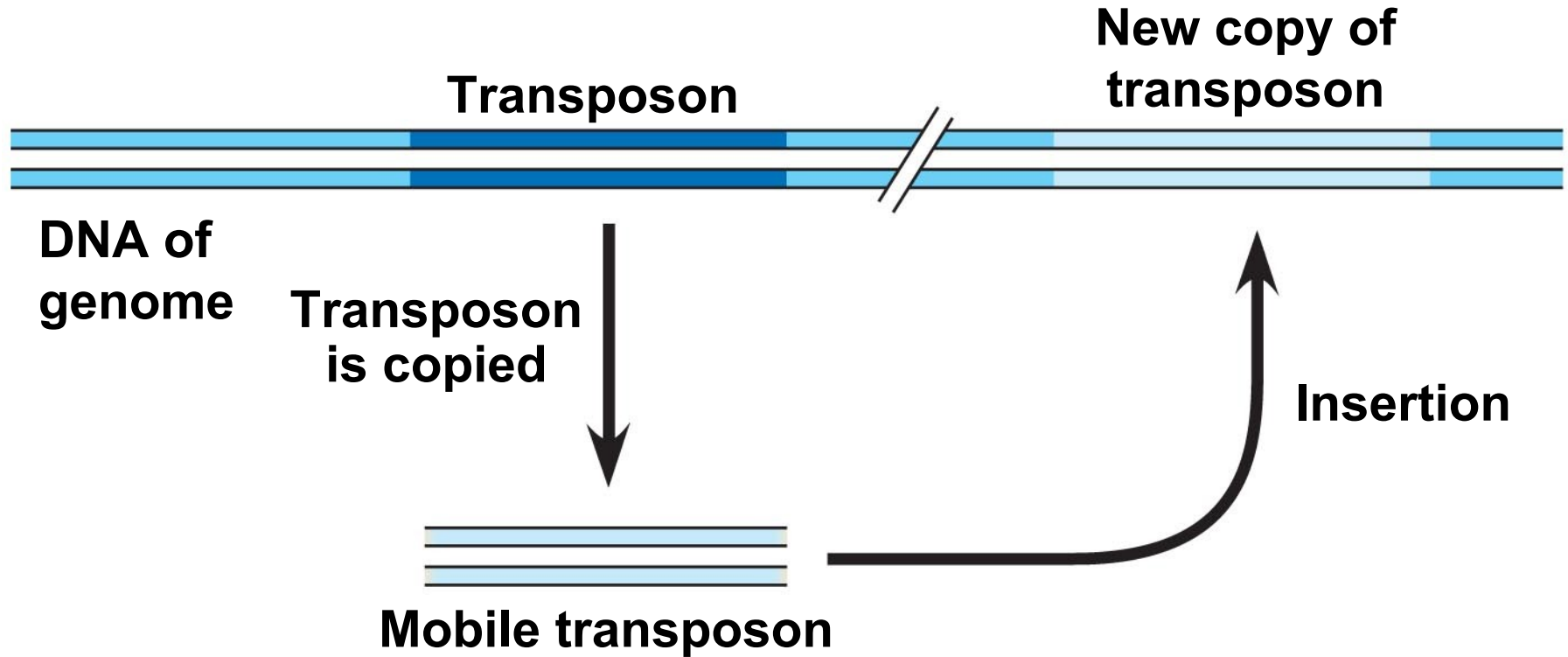
- Eukaryotic transposable elements are of two types:
 - **Transposons**, which move within a genome by means of a DNA intermediate
 - **Retrotransposons**, which move by means of an RNA intermediate



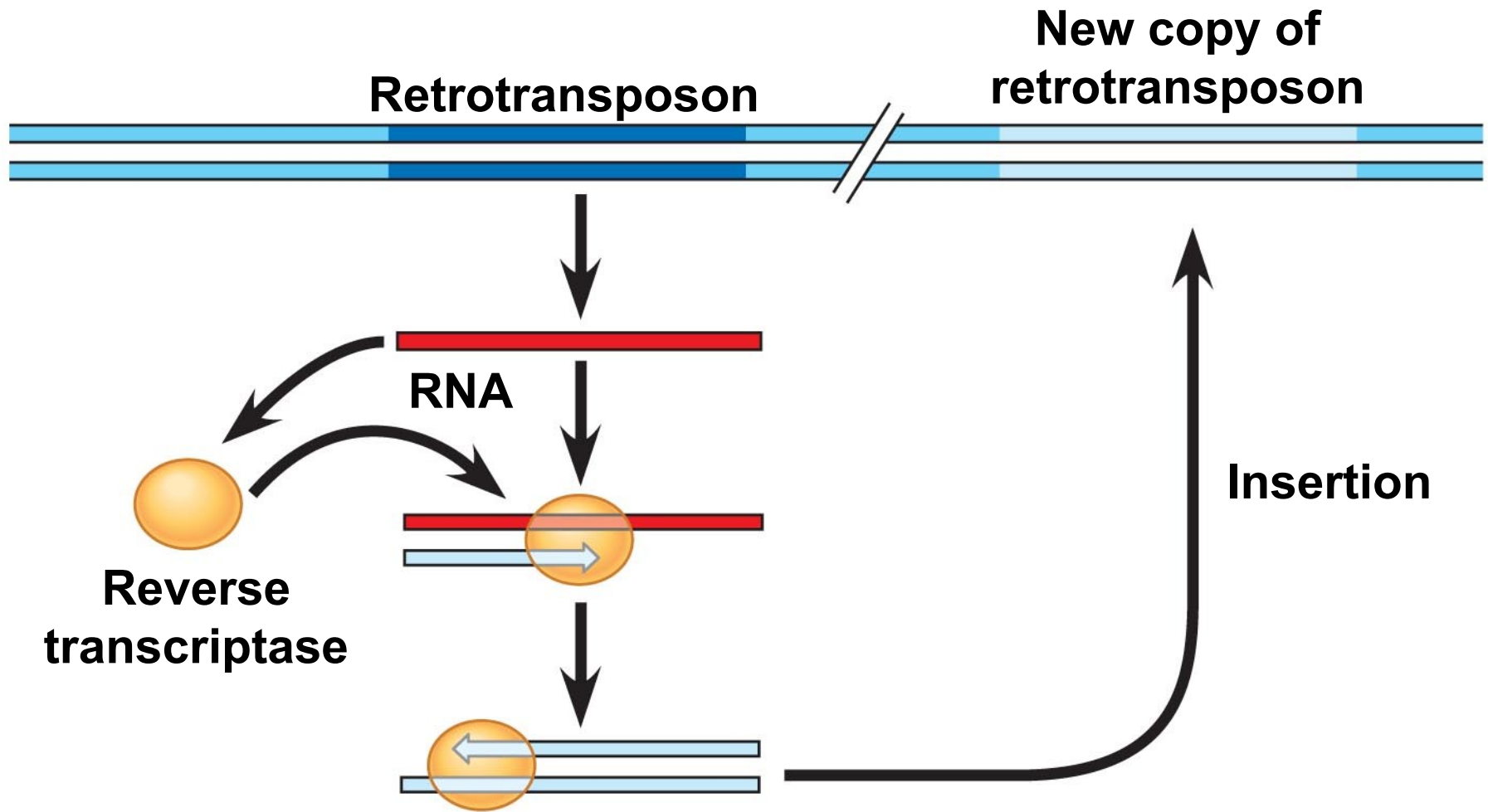
(a) Transposon movement ("copy-and-paste" mechanism)



(b) Retrotransposon movement



(a) Transposon movement (“copy-and-paste” mechanism)



(b) Retrotransposon movement

Sequences Related to Transposable Elements

- Multiple copies of transposable elements and related sequences are scattered throughout the eukaryotic genome
- In primates, a large portion of transposable element–related DNA consists of a family of similar sequences called *Alu elements*
- Many *Alu* elements are transcribed into RNA molecules; however, their function is unknown

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- The human genome also contains many sequences of a type of retrotransposon called *LINE-1* (*L1*)
 - L1 sequences have a low rate of transposition and may help regulate gene expression

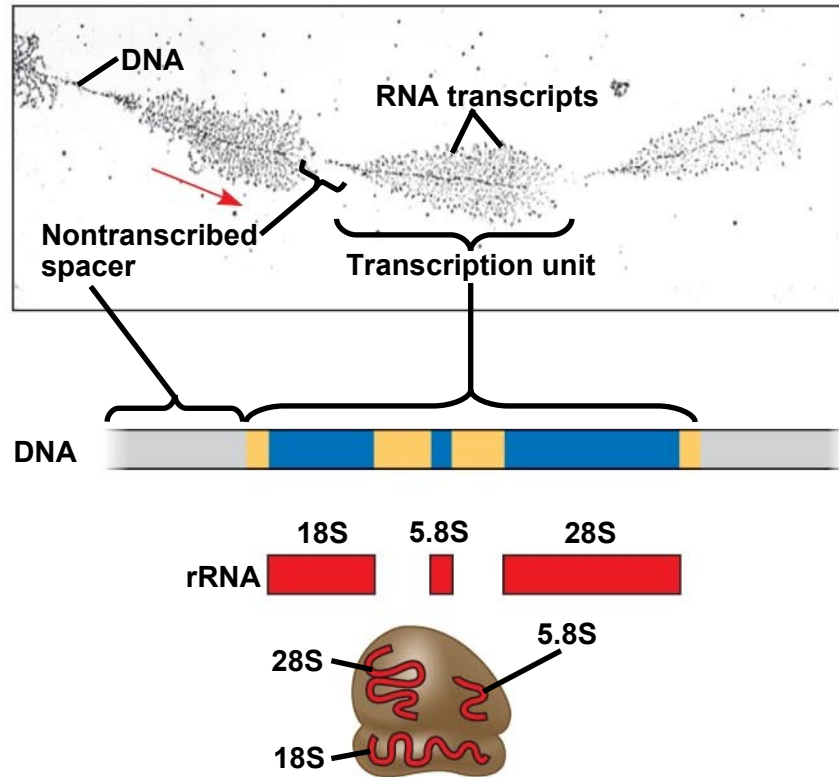
Other Repetitive DNA, Including Simple Sequence DNA

- About 15% of the human genome consists of duplication of long sequences of DNA from one location to another
- In contrast, **simple sequence DNA** contains many copies of tandemly repeated short sequences

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- A series of repeating units of 2 to 5 nucleotides is called a **short tandem repeat (STR)**
 - The repeat number for STRs can vary among sites (within a genome) or individuals
 - Simple sequence DNA is common in centromeres and telomeres, where it probably plays structural roles in the chromosome

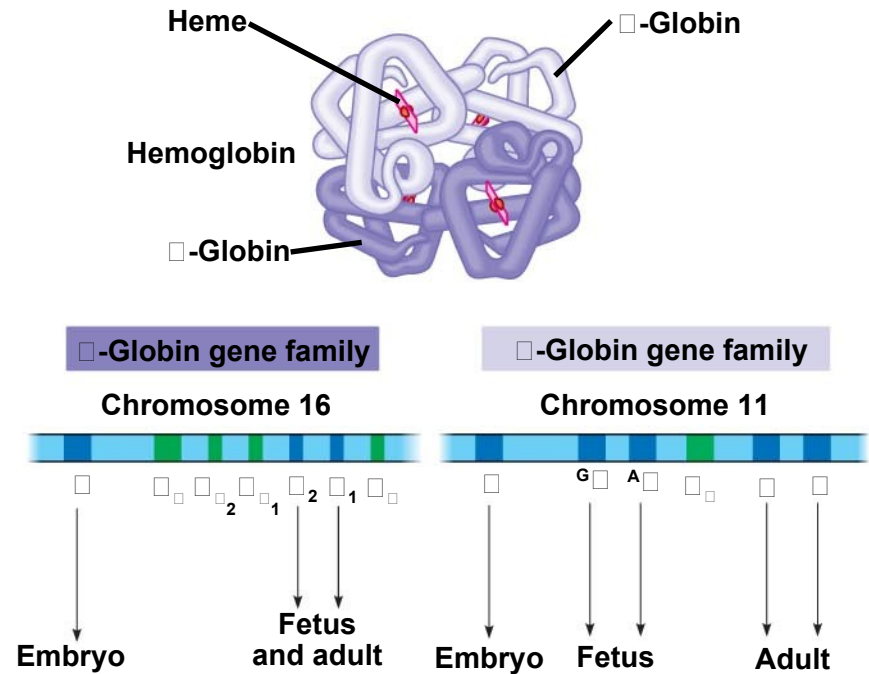
Genes and Multigene Families

- Many eukaryotic genes are present in one copy per haploid set of chromosomes
- The rest of the genome occurs in **multigene families**, collections of identical or very similar genes
- Some multigene families consist of identical DNA sequences, usually clustered tandemly, such as those that code for RNA products



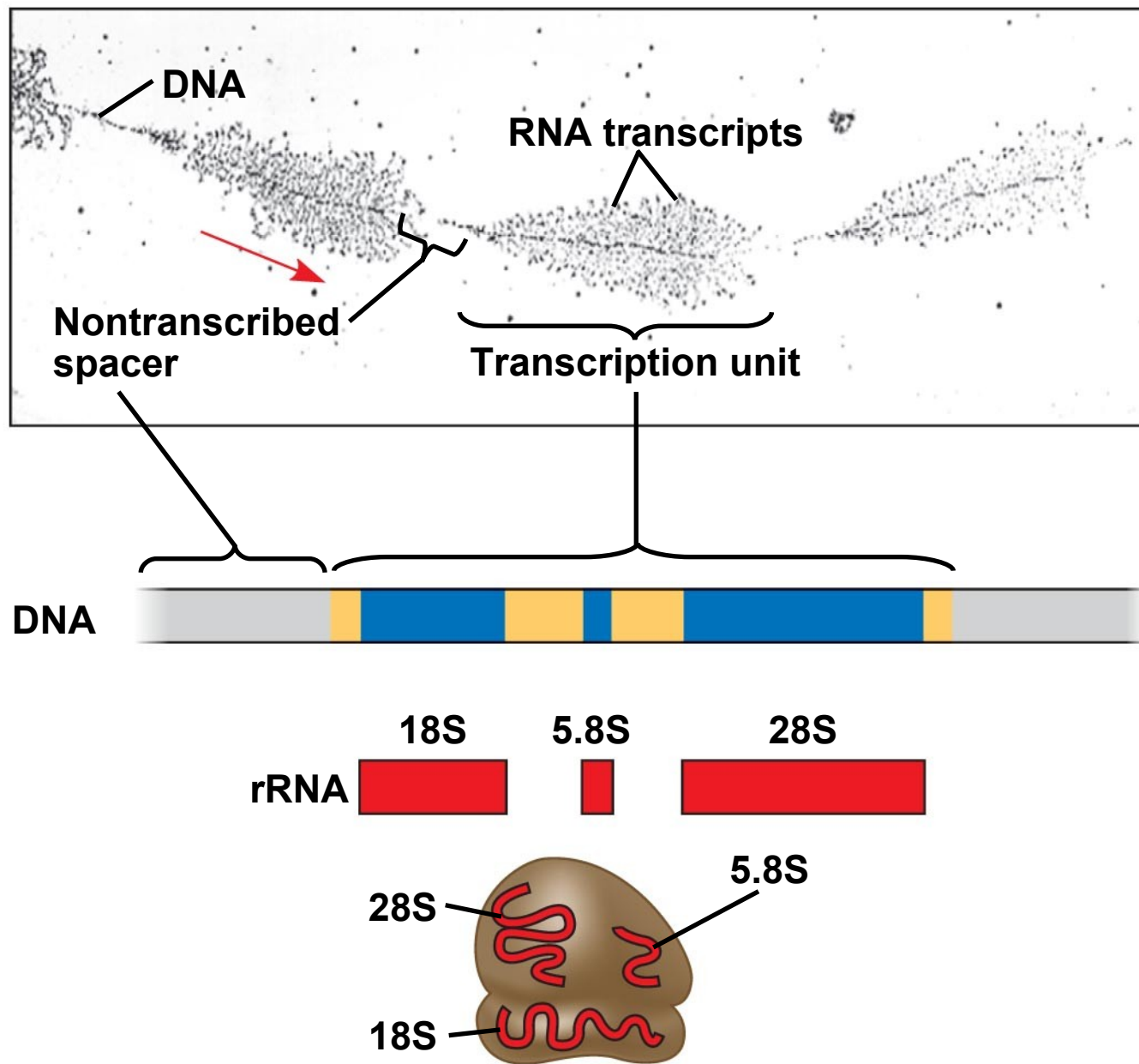
(a) Part of the ribosomal RNA gene family

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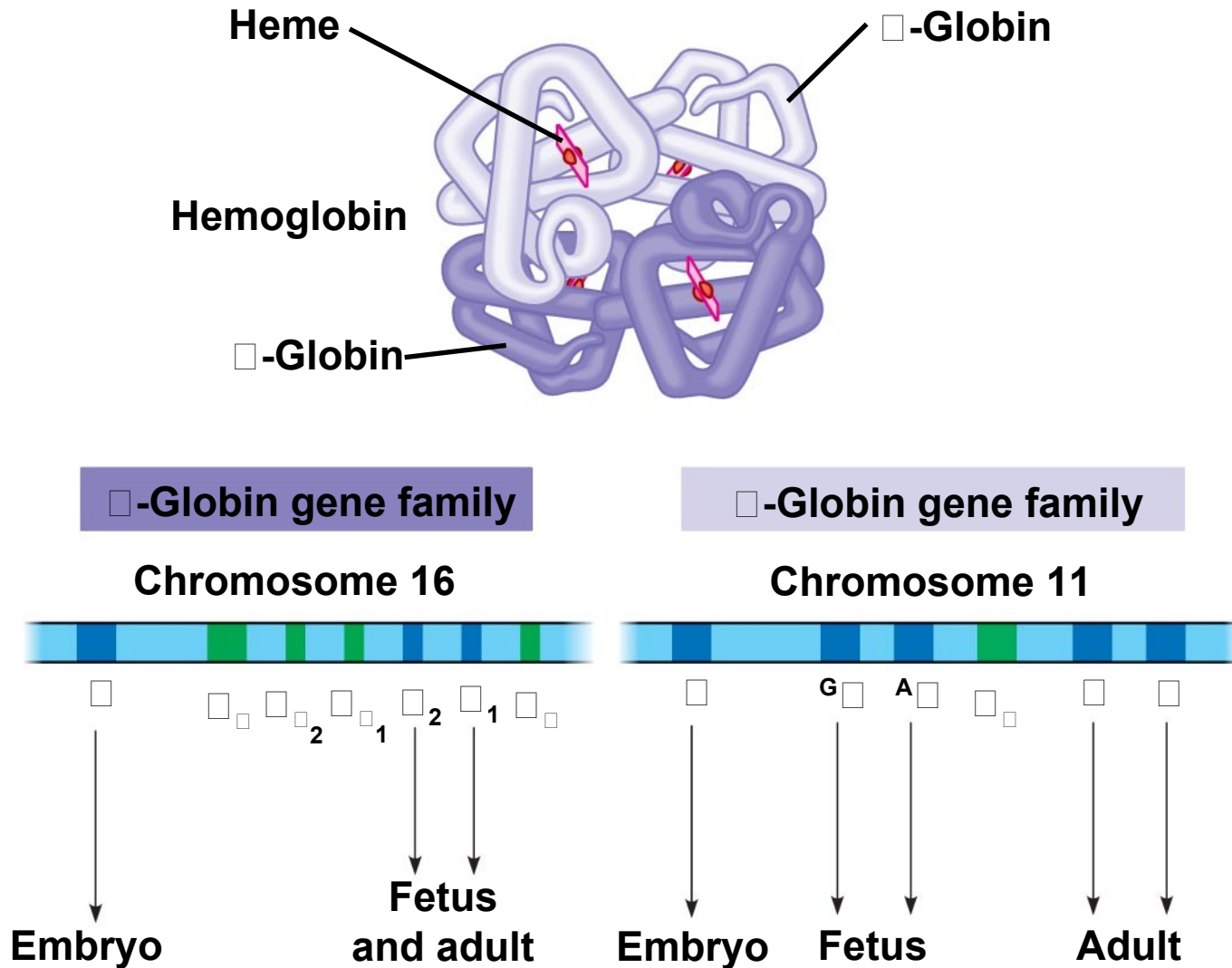
(b) The human α -globin and β -globin gene families

Fig. 21-10a



(a) Part of the ribosomal RNA gene family

-
- The classic examples of multigene families of nonidentical genes are two related families of genes that encode globins
 - α -globins and β -globins are polypeptides of hemoglobin and are coded by genes on different human chromosomes



(b) The human α -globin and β -globin gene families

Concept 21.5: Duplication, rearrangement, and mutation of DNA contribute to genome evolution

- The basis of change at the genomic level is mutation, which underlies much of genome evolution
- The earliest forms of life likely had a minimal number of genes, including only those necessary for survival and reproduction
- The size of genomes has increased over evolutionary time, with the extra genetic material providing raw material for gene diversification

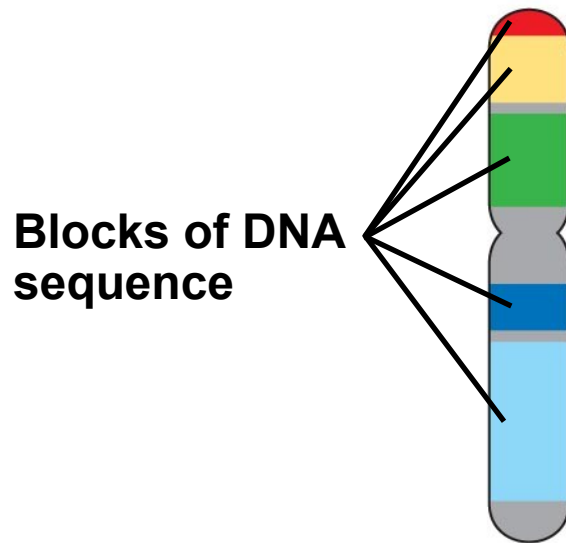
Duplication of Entire Chromosome Sets

- Accidents in meiosis can lead to one or more extra sets of chromosomes, a condition known as polyploidy
- The genes in one or more of the extra sets can diverge by accumulating mutations; these variations may persist if the organism carrying them survives and reproduces

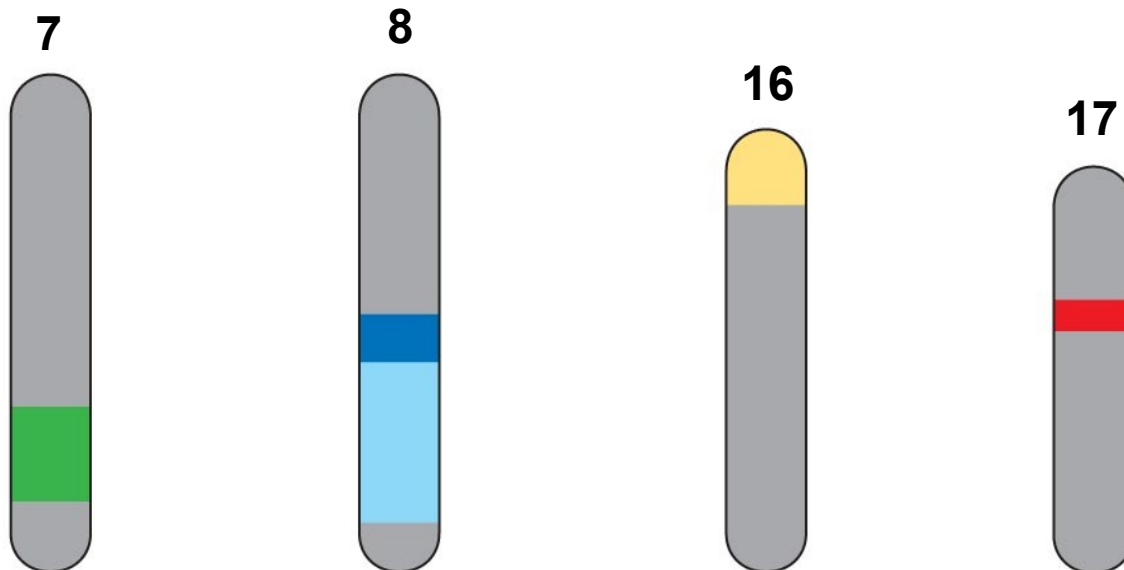
Alterations of Chromosome Structure

- Humans have 23 pairs of chromosomes, while chimpanzees have 24 pairs
- Following the divergence of humans and chimpanzees from a common ancestor, two ancestral chromosomes fused in the human line
- Duplications and inversions result from mistakes during meiotic recombination
- Comparative analysis between chromosomes of humans and 7 mammalian species paints a hypothetical chromosomal evolutionary history

Human chromosome 16



Blocks of similar sequences in four mouse chromosomes:

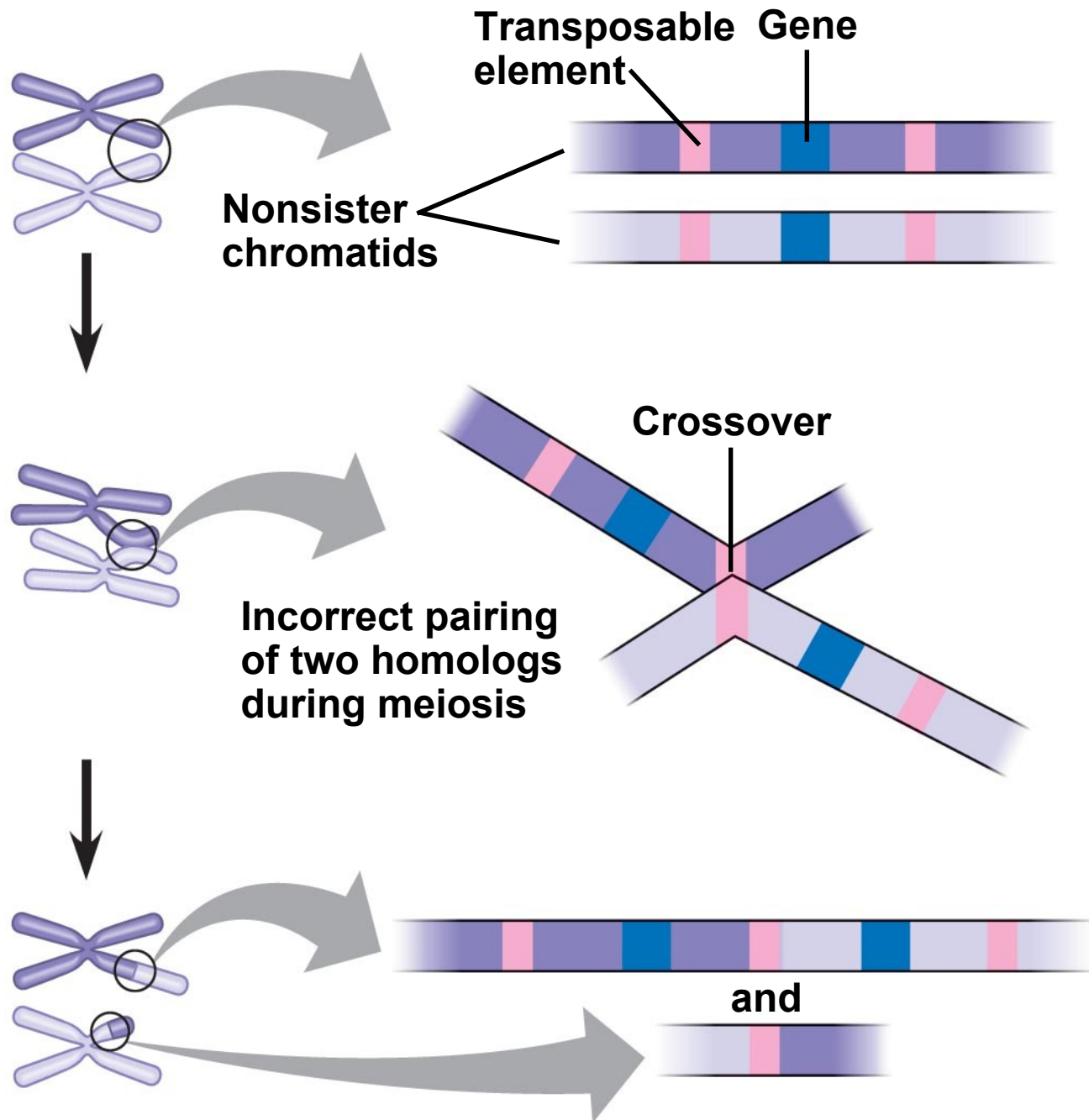


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- The rate of duplications and inversions seems to have accelerated about 100 million years ago
 - This coincides with when large dinosaurs went extinct and mammals diversified
 - Chromosomal rearrangements are thought to contribute to the generation of new species
 - Some of the recombination “hot spots” associated with chromosomal rearrangement are also locations that are associated with diseases

Duplication and Divergence of Gene-Sized Regions of DNA

- Unequal crossing over during prophase I of meiosis can result in one chromosome with a deletion and another with a duplication of a particular region
- Transposable elements can provide sites for crossover between nonsister chromatids

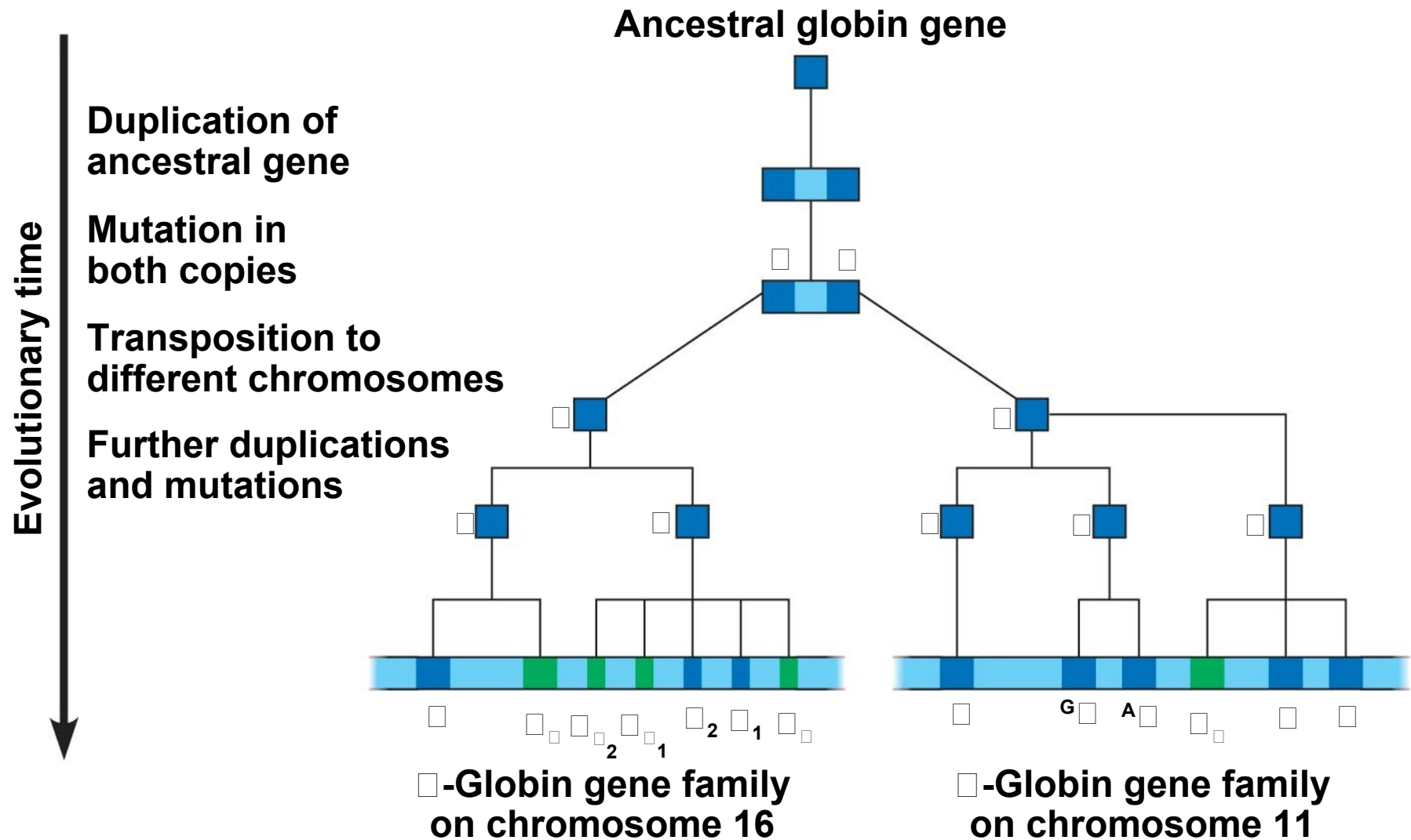
Fig. 21-12



Evolution of Genes with Related Functions: The Human Globin Genes

- The genes encoding the various globin proteins evolved from one common ancestral globin gene, which duplicated and diverged about 450–500 million years ago
- After the duplication events, differences between the genes in the globin family arose from the accumulation of mutations

Fig. 21-13



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- Subsequent duplications of these genes and random mutations gave rise to the present globin genes, which code for oxygen-binding proteins
 - The similarity in the amino acid sequences of the various globin proteins supports this model of gene duplication and mutation

Table 21.2 Percentage of Similarity in Amino Acid Sequence Between Human Globin Proteins

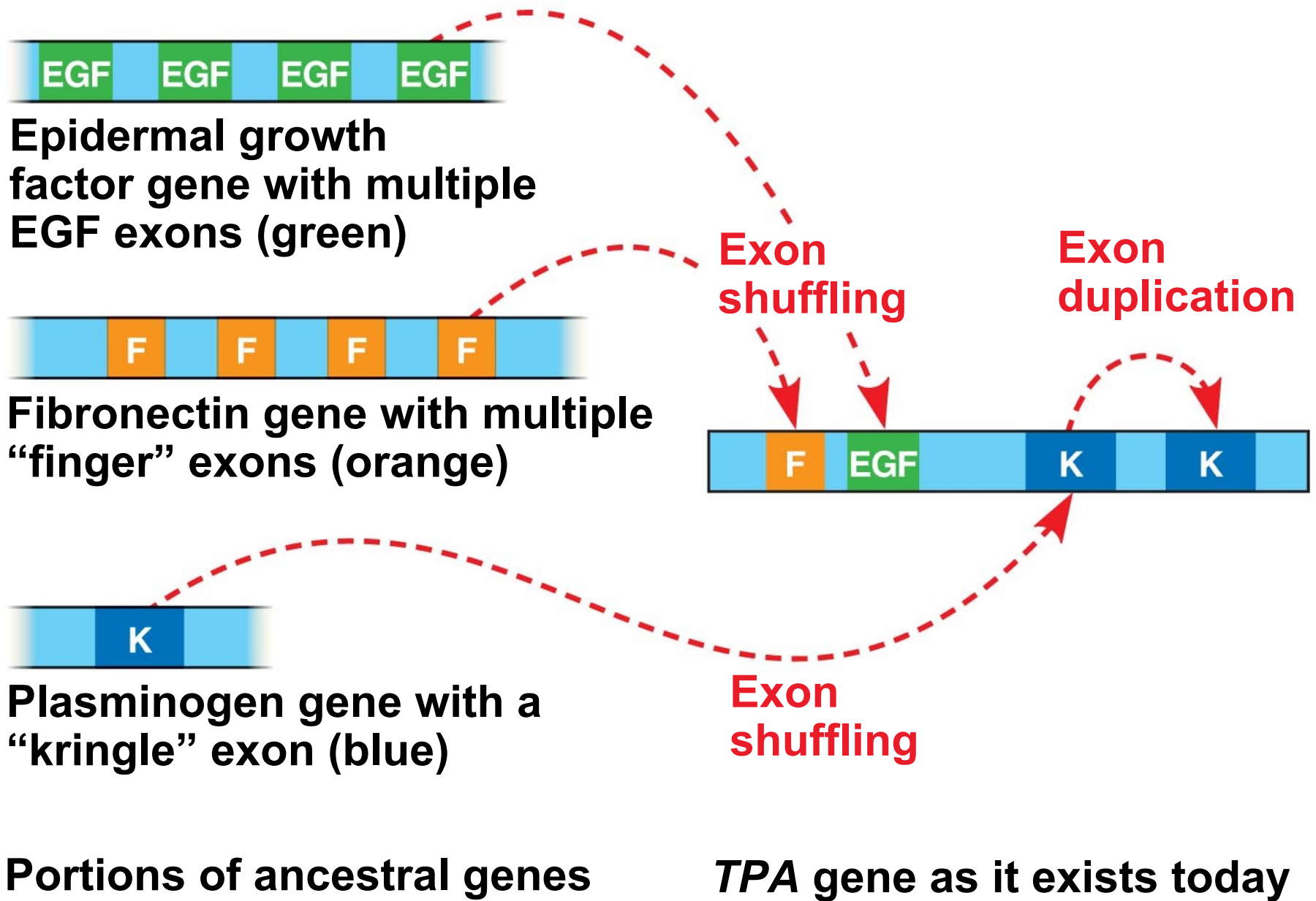
		α-Globins		β-Globins		
		α	ζ	β	γ	ϵ
α-Globins	α	100	58	42	39	37
	ζ	58	100	34	38	37
β-Globins	β	42	34	100	73	75
	γ	39	38	73	100	80
	ϵ	37	37	75	80	100

Evolution of Genes with Novel Functions

- The copies of some duplicated genes have diverged so much in evolution that the functions of their encoded proteins are now very different
- For example the lysozyme gene was duplicated and evolved into the α -lactalbumin gene in mammals
- Lysozyme is an enzyme that helps protect animals against bacterial infection
- α -lactalbumin is a nonenzymatic protein that plays a role in milk production in mammals

Rearrangements of Parts of Genes: Exon Duplication and Exon Shuffling

- The duplication or repositioning of exons has contributed to genome evolution
- Errors in meiosis can result in an exon being duplicated on one chromosome and deleted from the homologous chromosome
- In exon shuffling, errors in meiotic recombination lead to some mixing and matching of exons, either within a gene or between two nonallelic genes



How Transposable Elements Contribute to Genome Evolution

- Multiple copies of similar transposable elements may facilitate recombination, or crossing over, between different chromosomes
- Insertion of transposable elements within a protein-coding sequence may block protein production
- Insertion of transposable elements within a regulatory sequence may increase or decrease protein production

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- Transposable elements may carry a gene or groups of genes to a new location
 - Transposable elements may also create new sites for alternative splicing in an RNA transcript
 - In all cases, changes are usually detrimental but may on occasion prove advantageous to an organism

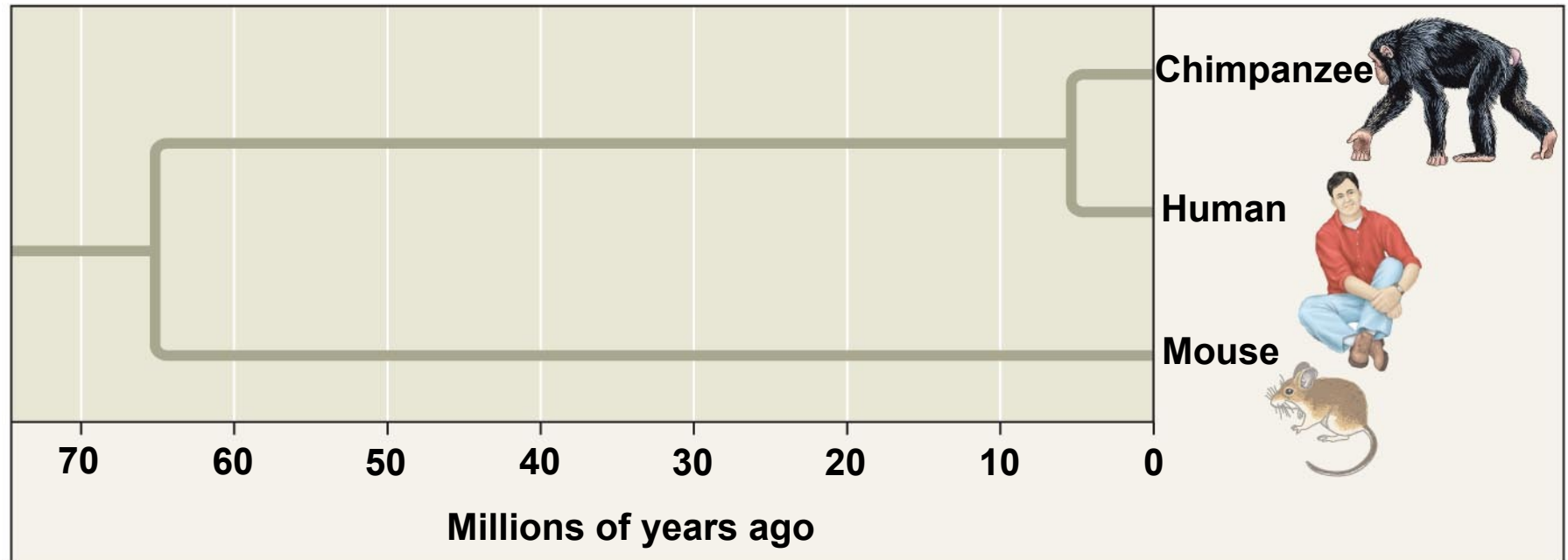
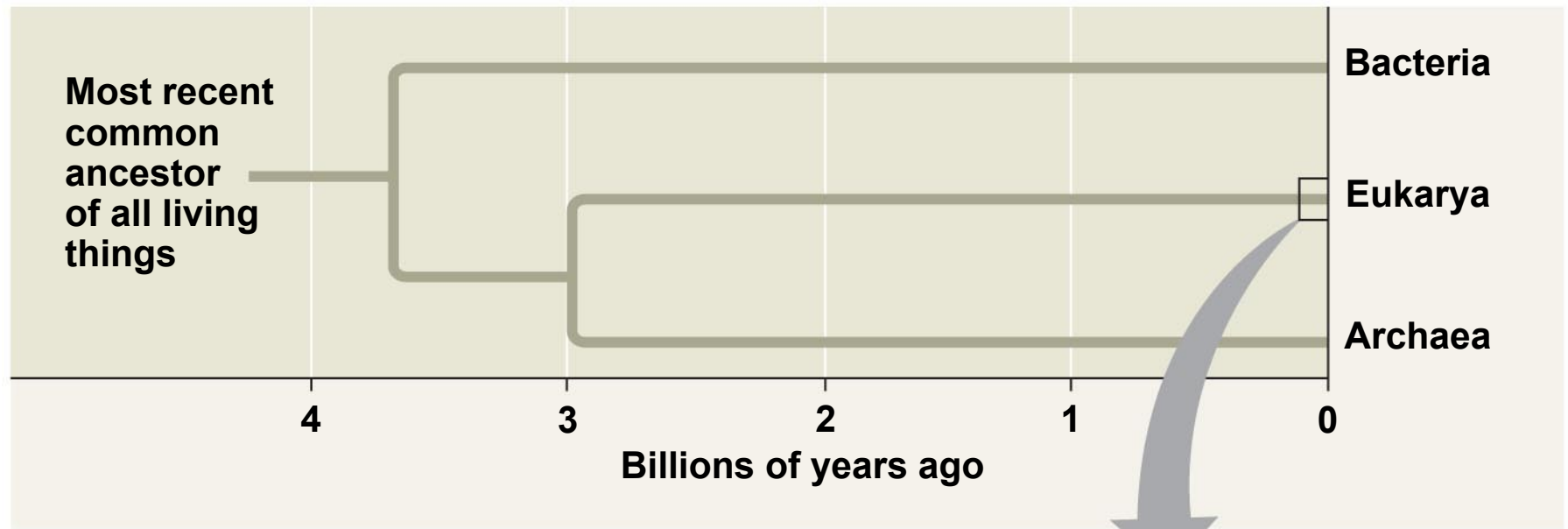
Concept 21.6: Comparing genome sequences provides clues to evolution and development

- Genome sequencing has advanced rapidly in the last 20 years
- Comparative studies of genomes
 - Advance our understanding of the evolutionary history of life
 - Help explain how the evolution of development leads to morphological diversity

Comparing Genomes

- Genome comparisons of closely related species help us understand recent evolutionary events
- Genome comparisons of distantly related species help us understand ancient evolutionary events
- Relationships among species can be represented by a tree-shaped diagram

Fig. 21-15



Comparing Distantly Related Species

- Highly conserved genes are genes that have changed very little over time
- These inform us about relationships among species that diverged from each other a long time ago
- Bacteria, archaea, and eukaryotes diverged from each other between 2 and 4 billion years ago
- Highly conserved genes can be studied in one model organism, and the results applied to other organisms

Comparing Closely Related Species

- Genetic differences between closely related species can be correlated with phenotypic differences
- For example, genetic comparison of several mammals with nonmammals helps identify what it takes to make a mammal

-
- Human and chimpanzee genomes differ by 1.2%, at single base-pairs, and by 2.7% because of insertions and deletions
 - Several genes are evolving faster in humans than chimpanzees
 - These include genes involved in defense against malaria and tuberculosis, regulation of brain size, and genes that code for transcription factors

-
- Humans and chimpanzees differ in the expression of the *FOXP2* gene whose product turns on genes involved in vocalization
 - Differences in the *FOXP2* gene may explain why humans but not chimpanzees communicate by speech

EXPERIMENT

Wild type: two normal copies of *FOXP2*

Heterozygote: one copy of *FOXP2* disrupted

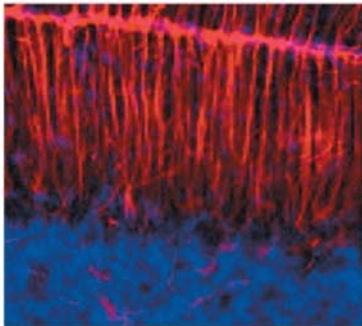
Homozygote: both copies of *FOXP2* disrupted

Experiment 1: Researchers cut thin sections of brain and stained them with reagents, allowing visualization of brain anatomy in a UV fluorescence microscope.

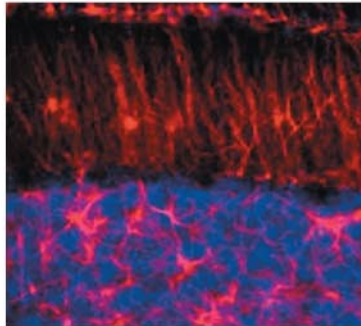
Experiment 2: Researchers separated each newborn pup from its mother and recorded the number of ultrasonic whistles produced by the pup.

RESULTS

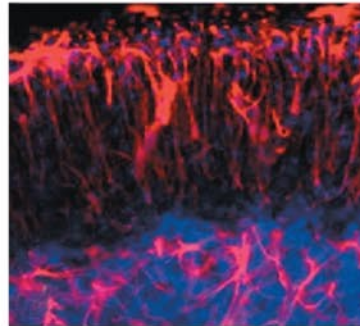
Experiment 1



Wild type

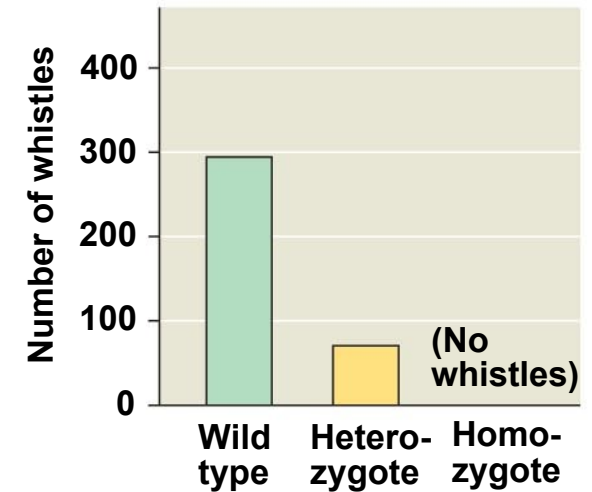


Heterozygote



Homozygote

Experiment 2



EXPERIMENT

Wild type: two normal copies of *FOXP2*

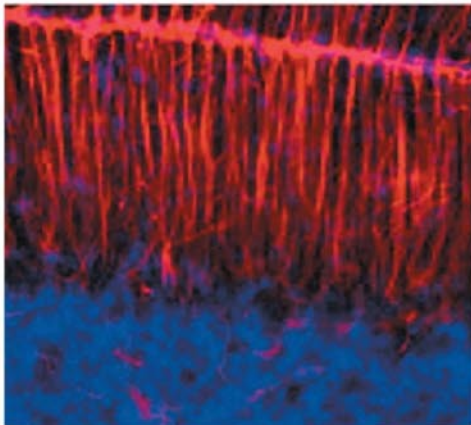
Heterozygote: one copy of *FOXP2* disrupted

Homozygote: both copies of *FOXP2* disrupted

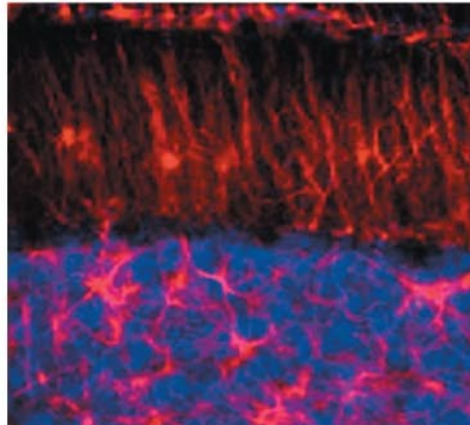
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RESULTS

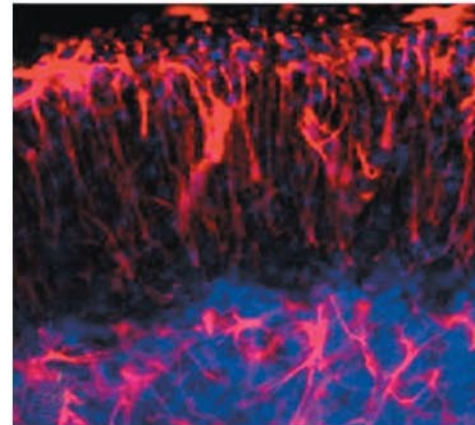
Experiment 1



Wild type



Heterozygote



Homozygote

EXPERIMENT

Wild type: two normal copies of *FOXP2*

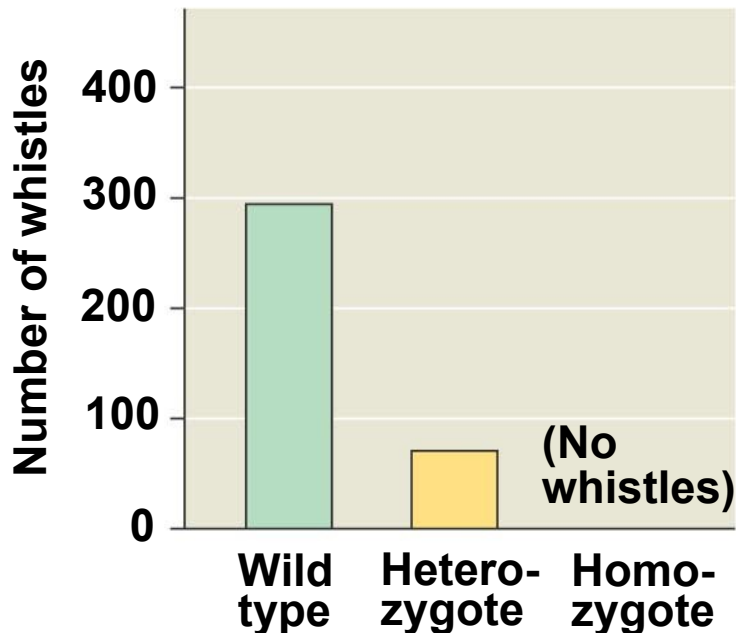
Heterozygote: one copy of *FOXP2* disrupted

Homozygote: both copies of *FOXP2* disrupted

Experiment 2: Researchers separated each newborn pup from its mother and recorded the number of ultrasonic whistles produced by the pup.

RESULTS

Experiment 2



Comparing Genomes Within a Species

- As a species, humans have only been around about 200,000 years and have low within-species genetic variation
- Variation within humans is due to single nucleotide polymorphisms, inversions, deletions, and duplications
- These variations are useful for studying human evolution and human health

Comparing Developmental Processes

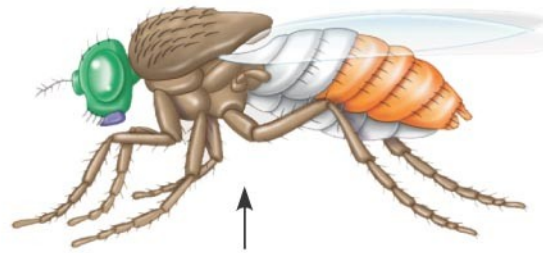
- Evolutionary developmental biology, or **evo-devo**, is the study of the evolution of developmental processes in multicellular organisms
- Genomic information shows that minor differences in gene sequence or regulation can result in major differences in form

Widespread Conservation of Developmental Genes Among Animals

- Molecular analysis of the homeotic genes in *Drosophila* has shown that they all include a sequence called a **homeobox**
- An identical or very similar nucleotide sequence has been discovered in the homeotic genes of both vertebrates and invertebrates
- Homeobox genes code for a domain that allows a protein to bind to DNA and to function as a transcription regulator
- Homeotic genes in animals are called *Hox* genes

Fig. 21-17

**Adult
fruit fly**



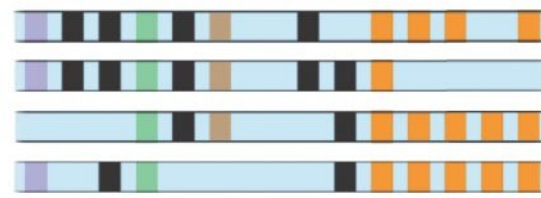
**Fruit fly embryo
(10 hours)**



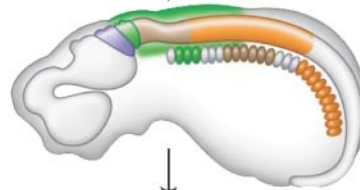
**Fly
chromosome**



**Mouse
chromosomes**



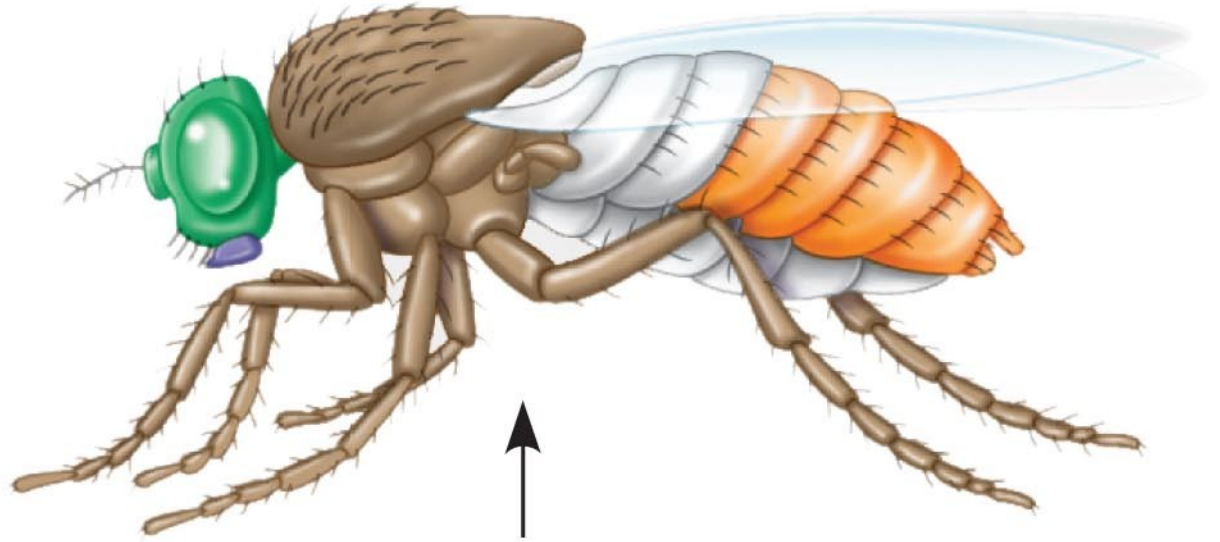
**Mouse embryo
(12 days)**



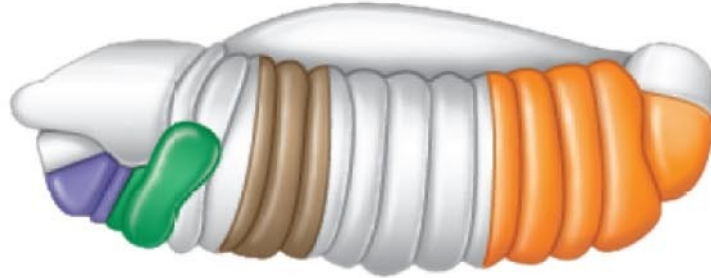
Adult mouse



**Adult
fruit fly**



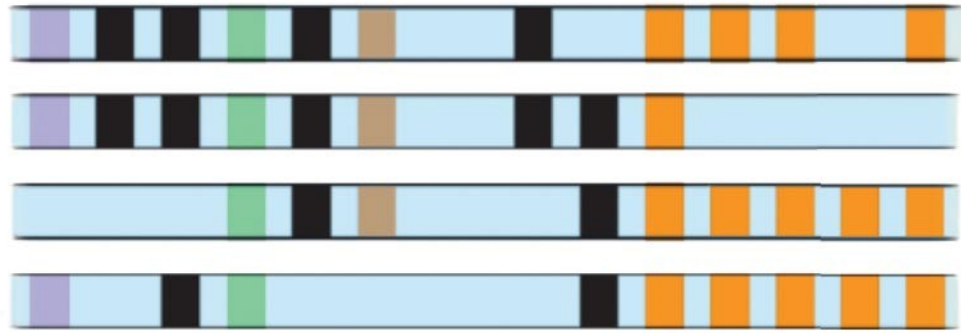
**Fruit fly embryo
(10 hours)**



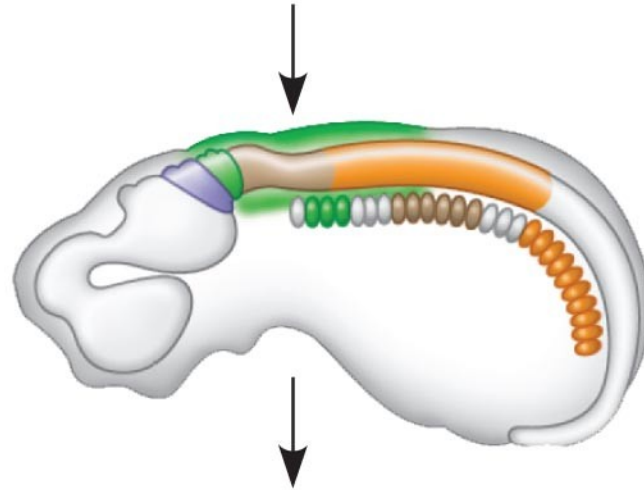
**Fly
chromosome**



**Mouse
chromosomes**



**Mouse embryo
(12 days)**

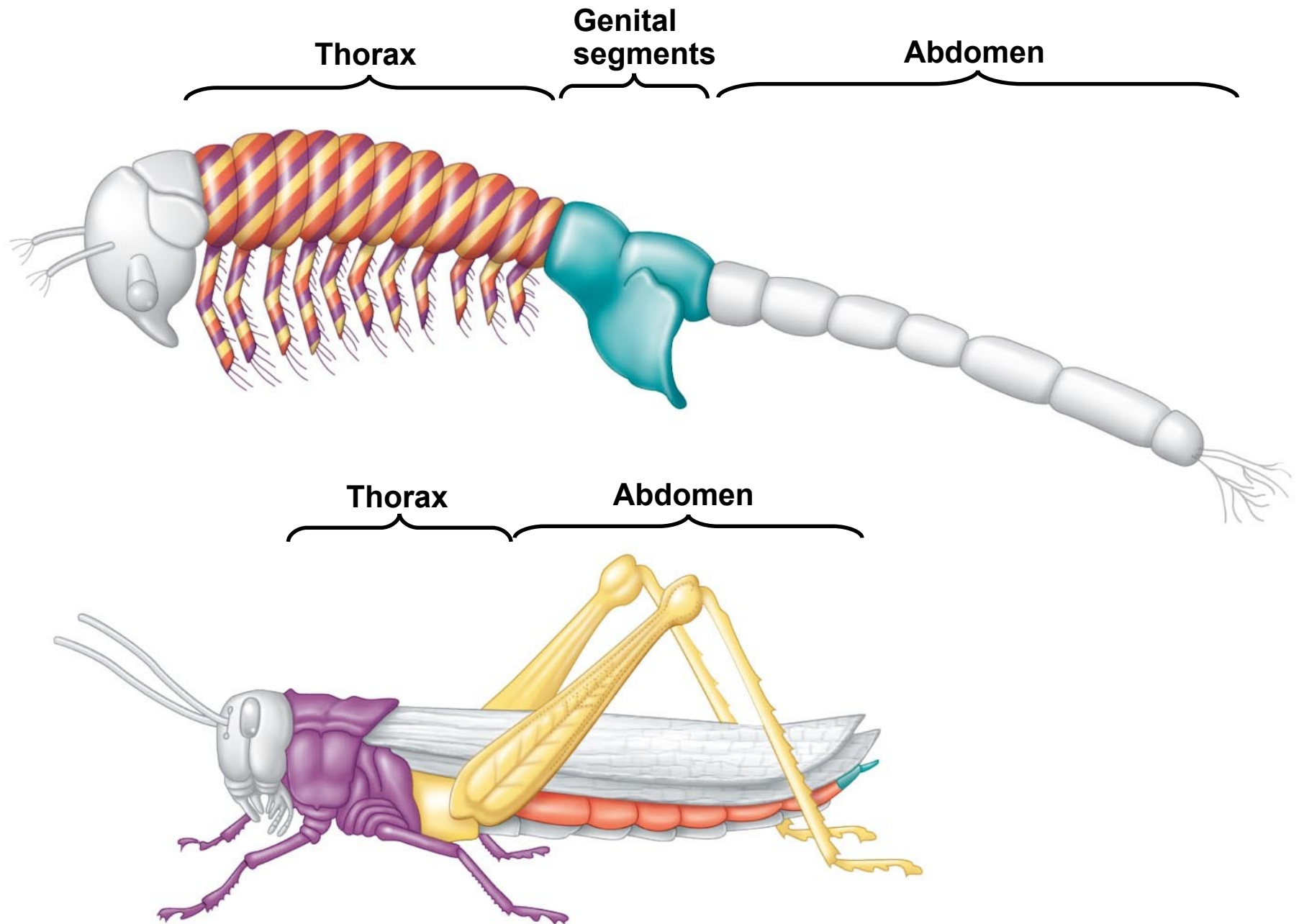


Adult mouse



-
- Related homeobox sequences have been found in regulatory genes of yeasts, plants, and even prokaryotes
 - In addition to homeotic genes, many other developmental genes are highly conserved from species to species

Fig. 21-18



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- Sometimes small changes in regulatory sequences of certain genes lead to major changes in body form
 - For example, variation in *Hox* gene expression controls variation in leg-bearing segments of crustaceans and insects
 - In other cases, genes with conserved sequences play different roles in different species

Comparison of Animal and Plant Development

- In both plants and animals, development relies on a cascade of transcriptional regulators turning genes on or off in a finely tuned series
- Molecular evidence supports the separate evolution of developmental programs in plants and animals
- *Mads-box* genes in plants are the regulatory equivalent of *Hox* genes in animals

Fig. 21-UN1

	Bacteria	Archaea	Eukarya
Genome size	Most are 1–6 Mb		Most are 10–4,000 Mb, but a few are much larger
Number of genes	1,500–7,500		5,000–40,000
Gene density	Higher than in eukaryotes		Lower than in prokaryotes (Within eukaryotes, lower density is correlated with larger genomes.)
Introns	None in protein-coding genes	Present in some genes	Unicellular eukaryotes: present, but prevalent only in some species Multicellular eukaryotes: present in most genes
Other noncoding DNA	Very little		Can be large amounts; generally more repetitive noncoding DNA in multicellular eukaryotes

1. **ATETI...PKSSD...TSSTT...NARRD**
2. **ATETI...PKSSE...TSSTT...NARRD**
3. **ATETI...PKSSD...TSSTT...NARRD**
4. **ATETI...PKSSD...TSSNT...SARRD**
5. **ATETI...PKSSD...TSSTT...NARRD**
6. **VTETI...PKSSD...TSSTT...NARRD**

1. ATETI...PKSSD...TSSTT...NARRD
2. ATETI...PKSSE...TSSTT...NARRD
3. ATETI...PKSSD...TSSTT...NARRD
4. ATETI...PKSSD...TSSNT...SARRD
5. ATETI...PKSSD...TSSTT...NARRD
6. VTETI...PKSSD...TSSTT...NARRD

You should now be able to:

1. Explain how linkage mapping, physical mapping, and DNA sequencing each contributed to the Human Genome Project
2. Define and compare the fields of proteomics and genomics
3. Describe the surprising findings of the Human Genome Project with respect to the size of the human genome
4. Distinguish between transposons and retrotransposons

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5. Explain how polyploidy may facilitate gene evolution
 6. Describe in general terms the events that may have led to evolution of the globin superfamily
 7. Explain the significance of the rapid evolution of the *FOXP2* gene in the human lineage
 8. Provide evidence that suggests that the homeobox DNA sequence evolved very early in the history of life